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Enhancing anaerobic digestion of sulfate-containing organic wastewater through an innovative integration of nitrate addition and microaeration

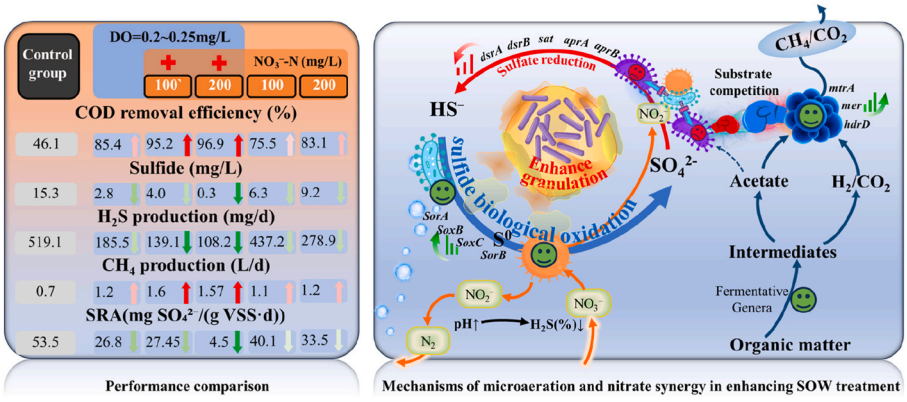
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HIGHLIGHTS

- A synergistic operational strategy combining nitrate addition and microaeration is proposed.
- Combined approach inhibits sulfate reduction, boosting methane production.
- Nitrate addition and microaeration enhances EPS secretion and sludge granulation.
- Microbial community dynamics and metabolic pathways drives enhanced SOW treatment.

GRAPHICAL ABSTRACT



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ABSTRACT

The reduction of sulfate to sulfide is a critical factor limiting the efficiency of anaerobic digestion in treating sulfate-containing organic wastewater. This study proposes a synergistic operational strategy that combines nitrate addition and microaeration to inhibit sulfate reduction and enhance anaerobic digestion performance. The combined application of nitrate (100–200 mg NO_3^--N/L) and microaeration (Dissolved oxygen: 0.20–0.25 mg/L) significantly promoted extracellular polymeric substance secretion and facilitated sludge granulation. Compared to the control group, which was operated without microaeration or nitrate addition, the combined strategy improved chemical oxygen demand removal efficiency and methane production by 48.2–49.8 % and 138.5–141.5 %, respectively, surpassing the enhancements achieved by nitrate addition alone (28.5–36.0 % and 69.2–86.2 %) or microaeration alone (38.3 % and 87.7 %, respectively). This synergistic approach inhibited the

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proliferation of sulfate-reducing bacteria (*Desulfovibrio*), while enriching key functional genera, including fermentative bacteria (*Lentimicrobium* and *Longilinea*), sulfur-oxidizing bacteria (*Sulfurospirillum* and *Thauera*), and methanogens (*Methanosaeta* and *Methanobacterium*). Consequently, sulfate-reduction genes (*aprA*, *aprB*, *dsrA*, and *dsrB*) were markedly downregulated, whereas sulfur-oxidation genes (*SoxX*, *SoxY*, *SoxZ*, *SoxB*, *SoxC*, and *SorA*) and methanogenesis-related genes (*mtrA*, *hdrD* and *mer*) were significantly upregulated. These findings elucidate the microbial community dynamics and metabolic pathways driving the enhanced anaerobic digestion of sulfate-containing organic wastewater under nitrate and microaeration conditions. This study provides a novel and effective strategy for optimizing the anaerobic digestion of sulfate-containing organic wastewater with substantial implications for sulfide control in wastewater treatment systems.

1. Introduction

Sulfate-containing organic wastewater (SOW) is generated by various industrial sectors, including the petrochemical, pharmaceutical, textile dyeing, mining processing, and food processing industries (Cetecioglu et al., 2019; Jung et al., 2019; Zak et al., 2021). Directly discharging untreated SOW can severely disrupt aquatic ecosystems and threaten aquatic organisms and human health. Therefore, the effective treatment of SOW is of significant importance for environmental protection and sustainable development.

Biological treatment methods, particularly anaerobic digestion, are widely used to manage organic wastewater because of their operational simplicity and cost-effectiveness. Anaerobic digestion offers multiple advantages over aerobic treatment technologies, including reduced aeration requirements, lower operating costs, a high organic loading capacity, minimal sludge production, and generating secondary energy in the form of methane (Fang et al., 2019). However, the chemical oxygen demand (COD) removal efficiency in the treatment of SOW generally remains below 80 % (Deng et al., 2022; Li et al., 2022), primarily because of the proliferation of sulfate-reducing bacteria (SRB). These bacteria reduce sulfate to sulfides through dissimilatory sulfate reduction, leading to sulfide accumulation, inhibiting microbial activity, and reducing the overall efficiency of anaerobic digestion. Free hydrogen sulfide (H_2S), in particular, is highly toxic and capable of penetrating cell membranes, forming cross-links within polypeptide chains, altering protein structures, and inhibiting microbial metabolic functions (Shi et al., 2024). Previous studies have shown that methanogenic activity is significantly inhibited when free H_2S concentrations exceed 50 mg/L, resulting in a 50 % reduction in methane production (Chen et al., 2008; Gil-Garcia et al., 2018; Wang et al., 2020). Furthermore, SRB exhibits a stronger substrate affinity for hydrogen and acetate than methanogens, exacerbating competition for these substrates and further suppressing methane production (Lu et al., 2016).

Various strategies have been explored to mitigate the adverse effects of sulfate on anaerobic digestion, including chemical precipitation and iron salt-based methods. Chemical precipitation with barium salts, calcium salts, or ettringite has been used to remove sulfate prior to digestion (Larraguibel et al., 2020; Pratinthong et al., 2021). The use of iron salts, such as $FeCl_2$ and zero-valent iron, has been shown to mitigate sulfide toxicity by reacting with sulfides to form iron sulfides, thereby reducing their harmful effects on microbial communities in anaerobic systems (Juntupally et al., 2022; Sun et al., 2019). In particular, the precipitation and coagulation processes generate sludge that can be potentially toxic due to the accumulation of heavy metals and other hazardous substances. Therefore, the safe disposal or treatment of this sludge is essential to prevent environmental contamination and secondary risks (Zhao et al., 2024). In addition, operational adjustments, such as maintaining influent pH above 8.0 and optimizing the COD to sulfur ratio and implementing two-phase anaerobic digestion, have been explored to alleviate sulfide-induced inhibition (Hu et al., 2022b; Rajendran et al., 2020; Rodriguez et al., 2012). These interventions fail to completely eliminate sulfate reduction, and their effectiveness in enhancing anaerobic digestion remains limited. Thus, there is an urgent need to develop innovative and cost-effective strategies to mitigate sulfate reduction and enhance the anaerobic digestion of SOW.

Nitrate has been widely recognized as an effective inhibitor of sulfate reduction in anaerobic systems. Compared to other chemical additives, nitrate is not only readily available from industrial sources such as petrochemical, steel, and fertilizer production, but is also abundant in nitrification effluents from wastewater treatment plants (Abascal et al., 2022; Inaba et al., 2022). Harnessing wastewater-derived nitrate for SOW treatment offers the dual benefits of suppressing sulfide formation and implementing a circular "waste-to-treat-waste" strategy. The suppression of SRB by nitrate occurs primarily through competition for electron donors by nitrate-reducing bacteria (NRB), as well as the production of nitrite during denitrification, which further inhibits SRB activity (Lai et al., 2020; Greene et al., 2003; Zhao et al., 2009). Additionally, the enrichment of sulfur-oxidizing nitrate-reducing bacteria (SoNRB) promotes the oxidation of sulfides to sulfate or elemental sulfur, thereby alleviating the inhibitory effects of sulfides on methanogenesis (Cai et al., 2021). Experimental studies have demonstrated the effectiveness of nitrate addition in enhancing SOW treatment. For example, supplementation with 300 mg/L nitrate-nitrogen decreased sulfate removal from 80.5 % to 68.0 % and simultaneously increased COD removal from 63.3 % to 83.1 % (Liao et al., 2014). Besides nitrate, microaeration has also been shown to inhibit sulfate reduction and improve anaerobic digestion performance. Under controlled low-oxygen conditions, the activity of sulfide-oxidizing bacteria (SOB) is enhanced, facilitating the conversion of sulfide to less harmful sulfur species and markedly reducing H_2S emissions (Nguyen and Khanal, 2018). For instance, Zhou et al. (2007) demonstrated that increasing the air sparging rate from 0 to 6 mL/(L·min) during anaerobic treatment of pulp mill wastewater reduced the sulfide concentration from 400 to 200 mg/L, while increased COD removal by 40 % and methane production by 700 %. Similarly, elevating dissolved oxygen levels from 0.02 to 0.25 mg/L improved COD removal from 32.5 % to 69.3 % in anaerobic digestion systems (Yu et al., 2014). Although previous studies have examined the individual effects of nitrate addition and microaeration on the anaerobic digestion performance of SOW, most have primarily focused on digestion efficiency and microbial community composition. There has been limited exploration of the physicochemical characteristics of sludge and the distribution of functional genes under these conditions. However, few studies have systematically investigated the synergistic interaction between nitrate addition and microaeration. In particular, their combined influence on anaerobic digestion efficiency, sulfate reduction, and the associated mechanistic pathways remains poorly understood. From an engineering perspective, it is also necessary to determine the threshold concentration of nitrate under microaerobic conditions at which sulfate reduction begins to be inhibited, as well as to evaluate the feasibility of implementing such combined strategies in practical wastewater treatment systems.

To address these knowledge gaps, this study systematically investigated the synergistic effects of microaeration and nitrate addition on the anaerobic digestion of SOW. The impacts of varying nitrate concentrations under microaerobic conditions on pollutant removal efficiency, sludge physicochemical properties, microbial community structure, and the abundance of functional genes were comprehensively examined. This study not only elucidated the underlying mechanisms by which

microaeration and nitrate jointly inhibited sulfate reduction and enhanced treatment performance, but also proposed a novel and practical strategy for optimizing anaerobic processes and controlling sulfide generation in SOW treatment. These findings provided new insights into the development of advanced, sustainable, and resource-efficient technologies for the management of SOW, addressing a critical gap in current wastewater treatment research and practice.

2. Materials and methods

2.1. Seed sludge and synthetic SOW

Seed sludge was obtained from a wastewater treatment station at the Guangdong University of Petrochemical Technology, China. The volatile suspended solid (VSS) to total suspended solid (TSS) ratio was 0.55. The synthetic SOW was prepared according to the nutrient composition typically found in long-chain dicarboxylic acid (LCDA) wastewater from petrochemical industries. Due to its relatively simple composition, with glucose ($C_6H_{12}O_6$) as the primary carbon source, this approach minimized the potential confounding effects of more complex organic substrates, thereby enabling a more precise assessment of the impacts of the experimental variables (nitrate addition and microaeration) on microbial metabolism. The sulfate ions were supplied by sodium sulfate (Na_2SO_4). Ammonium chloride (NH_4Cl) and potassium dihydrogen phosphate (KH_2PO_4) were added as nitrogen and phosphorus sources, respectively, to achieve a COD:N:P ratio of approximately 200:5:1. Trace elements were added according to the protocol described by Chen et al. (2017). Sodium bicarbonate ($NaHCO_3$) was added as a buffering agent at concentrations ranging from 840 to 1344 mg/L to maintain the pH of wastewater within the range of 8.0–8.5.

2.2. Experimental setup and operation procedure

The experiment was conducted using three UASB reactors equipped with an external aeration device (Fig. 1). R1 served as the control group, with no nitrate addition or microaeration, for the first 135 days. R2 was the experimental group with both microaeration and nitrate addition, while R3 served as a control group with only nitrate addition. Each UASB reactor had an effective volume of 4.7 L, internal diameter of 74 mm, and a height of 1195 mm. The synthetic SOW was introduced into the UASB reactor via a peristaltic influent pump and transferred to an external aeration unit using a second peristaltic pump (recirculating pump 1) for aeration. The aerated wastewater was subsequently recirculated into the reactor along with the influent using a recirculating pump (recirculating pump 2). The gases released during aeration and anaerobic digestion were directed through a NaOH absorption bottle, which served to remove both H_2S and CO_2, thereby preventing their release into the laboratory environment. For periodic monitoring of methane and H_2S concentrations, the NaOH absorption bottle was temporarily removed, and the biogas was collected directly into gas sampling bags for subsequent analysis. In R2, the dissolved oxygen (DO) level was controlled by recirculating aerated wastewater from an external aeration device. The reflux ratio was maintained at 200 %, thereby keeping the DO concentration in the UASB was maintained within a range of 0.20–0.25 mg/L. Throughout the experimental period, the hydraulic retention time (HRT) was maintained at 24 h, and the operational temperature was controlled at 38 ± 1 °C using a water bath.

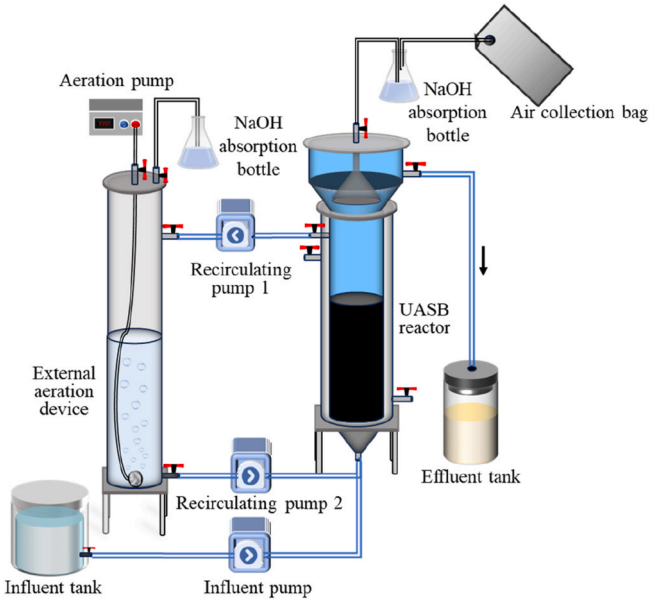


Fig. 1. Diagram of experimental setup.

The reactors were operated for 166 d. During the initial 12 d, the influent sulfate concentration in all three reactors was maintained at 400 mg/L, whereas the influent COD gradually increased from 500 to 1500 mg/L. From day 12 to day 166, the influent COD was set at 1500 mg/L, achieving a COD/SO_4^{2-} ratio of 3.75 to ensure sufficient carbon availability for sulfate reduction (Li et al., 2024a). To evaluate the effects of nitrate concentration and microaerobic conditions on anaerobic digestion for SOW treatment, nitrate was sequentially introduced into the influents of reactors R2 and R3 at concentrations of 50, 100, and 200 mg NO_3^--N/L on days 1, 76, and 113, respectively. These concentrations were chosen based on the typical range (50–300 mg NO_3^--N/L) reported in the literature to effectively suppress SRB and enhance anaerobic digestion (Liao et al., 2014; Cai et al., 2021), as well as the experimental goal of identifying the minimum nitrate dosage required to significantly improve treatment efficiency. To assess the effect of microaeration alone on anaerobic digestion during SOW treatment, aeration was performed in the external recirculation device of R1 from day 136 to day 166. During this phase, the reflux ratio was maintained at 200 %, and the DO concentration in the UASB reactor was controlled within the range of 0.20–0.25 mg/L, consistent with the operational parameters used for R2.

2.3. Analytical methods

The COD, TSS, VSS, and sludge volume index (SVI_{30}) were analyzed using standardized methods (APHA and AWWA, 2005). Nitrate, nitrite and sulfate concentrations were determined via ion chromatography (Thermo Fisher, ICS-600, USA), whereas sulfide concentrations were measured using the methylene blue method (Trüper and Schlegel, 1964). The pH and DO were measured with a portable pH meter (SMART SENSOR, PH 838, China) and a portable dissolved oxygen meter (SMART SENSOR, AT9210, China), respectively. The contents of methane (CH_4) and H_2S were determined using gas chromatography (Agilent Technologies 6890N, Santa Clara, CA, USA) and a hydrogen sulfide detection tube (Gastec, No.4H, China), respectively. The volumes of CH_4 and H_2S were determined using a 100 mL syringe, with the measurements recorded at a temperature of 298.15 K and a pressure of 101.325 kPa. The competitive relationship between SRB and methanogens for substrate utilization was quantitatively evaluated using the electron flow ratio of SRB, which was calculated according to Equation (1). A higher proportion of electron flow allocated to the SRB indicates a competitive advantage in substrate utilization over methanogens (Isa et al., 1986),

$$e = \frac{\Delta SO_4^{2-} \times 0.72}{\Delta COD} \times 100\% \quad (1)$$

where e represents the fraction of electron flow attributed to SRB, ΔCOD and ΔSO_4^{2-} represent the total amounts of COD and SO_4^{2-} removed, respectively.

The sludge particle size was measured using a wet sieving method

(Laguna et al., 1999). Extracellular polymeric substances (EPS) in the sludge were extracted using a thermal extraction method (Lv et al., 2014). Protein (PN) and polysaccharide (PS) contents were determined using the Lowry method (Lowry et al., 1951) and the phenol-sulfuric acid method (DuBois et al., 1956), respectively. The sulfate-reducing activity (SRA) of the sludge was determined following the procedure described by Lu et al. (2016). Specifically, 50 mL of synthetic SOW and 250 mg of anaerobic sludge were introduced into a 100 mL serum bottle equipped with a gas collection bag. The mixture was stirred in a water bath shaker at 38 °C and 180 rpm for 24 h. After filtration through a 0.45 μm membrane, the sulfate concentration was measured to evaluate the activity. All measurements were performed in triplicate, and the results are presented as mean ± standard deviation.

2.4. Microbial community analysis

Sludge samples were collected from the UASB reactors operating under varying conditions to analyze the microbial community structure. DNA was extracted using the MolPure Soil DNA Kit. The V4 region of the 16S rRNA gene was amplified with universal primers 515F (5′-GTGYCAGCMGCCGCGGTAA-3′) and 806R (5′-GGACTACNVGGGTWTCTAAT-3′) (Apprill et al., 2015; Parada et al., 2015). These primers are widely applied and can amplify both bacterial and archaeal sequences, including methanogens, in diverse non-marine environments (Klindworth et al., 2013; Walters et al., 2016). However, primer-dependent biases may result in the underrepresentation of archaeal groups, particularly specific methanogenic lineages. Thus, the reported methanogen abundance should be considered conservative. Sequencing data were processed and analyzed using the Majorbio I-Sanger Cloud platform (www.i-sanger.com), following the methodology outlined by Liang et al. (2019). Detailed protocols and conditions for DNA extraction, PCR and sequencing data processing are provided in the Supplementary Materials.

Functional gene profiles were predicted from 16S rRNA sequencing data using the PICRUSt pipeline, with functional annotation based on the latest KEGG database, as detailed by Langille et al. (2013). To evaluate the enrichment of a specific functional gene in each reactor relative to the sulfate-acclimated sludge (R1-135), the odds ratio (OR) was calculated using Equation (2), following the approach of Liang et al. (2019).

$$OR = \frac{\varphi_R/(1-\varphi_R)}{\varphi_0/(1-\varphi_0)} \quad (2)$$

where φ_R is the relative abundance of a specific metabolic category associated with a given gene in the UASB system, and φ_0 denotes the relative abundance of the same metabolic category associated with the gene in the sulfate-acclimated sludge.

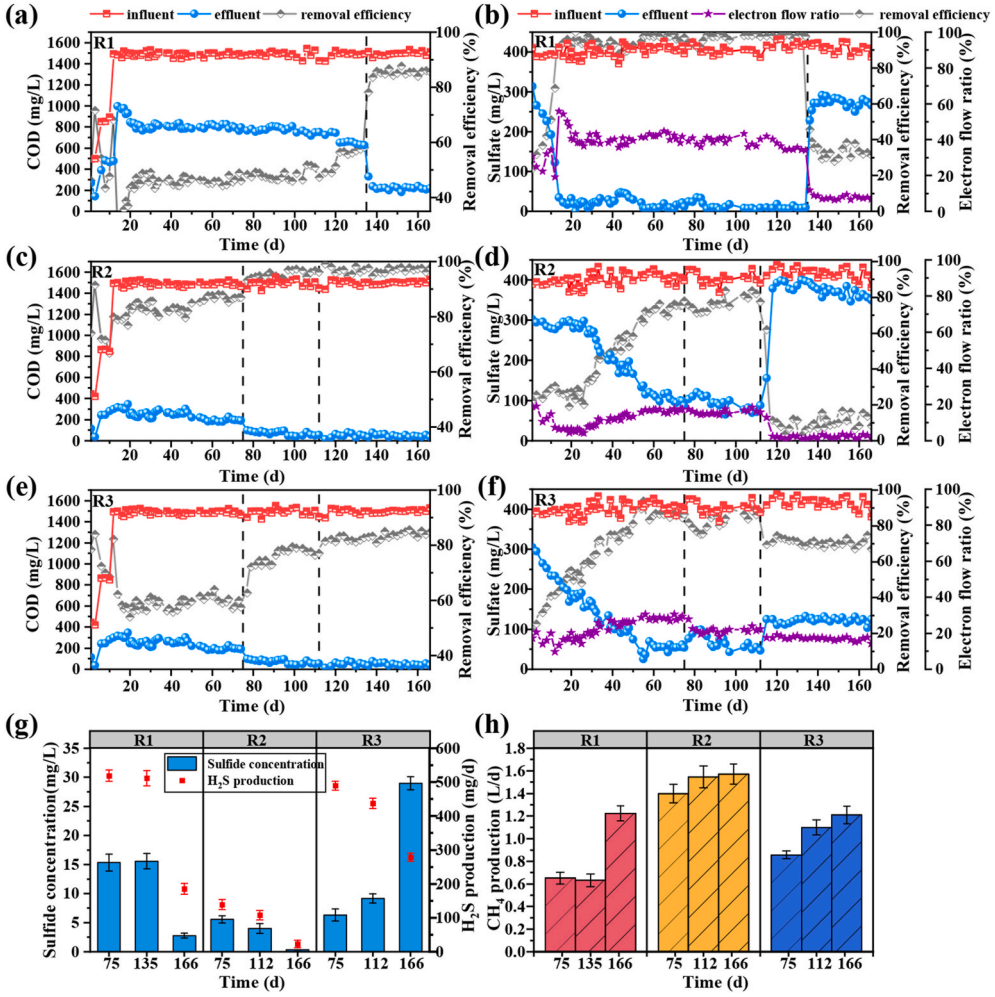


Fig. 2. Performance of UASB reactors over the experiment period: (a) COD removal in R1; (b) Sulfate removal and electron flow ratio in R1; (c) COD removal in R2; (d) Sulfate removal and electron flow ratio in R2; (e) COD removal in R3; (f) Sulfate removal and electron flow ratio in R3; (g) Sulfide concentration and hydrogen sulfide production over the experiment period; (h) Methane production over the experiment period.

3. Results and discussion

3.1. Reactors performance

The variations in COD removal, sulfate removal, and electron flow ratio across different reactors over the experimental period are shown in Fig. 2(a–f). In the control group R1, which was operated strictly under anaerobic conditions without nitrate addition or microaeration during the first 135 days, the average removal efficiencies of COD and sulfate, as well as the electron flow ratio, were 47.0 %, 96.7 %, and 39.5 %, respectively. This prolonged anaerobic operation ensured that R1 functioned as a stable control reactor prior to the introduction of microaeration. Microaeration was subsequently introduced to R1 on day 136, after which the effluent COD and sulfate concentrations stabilized within approximately 15 days. The rapid stabilization of these performance indicators suggests that R1 reached a functional steady state under microaeration conditions, enabling a robust evaluation of microaeration effects within this operational period. Correspondingly, the average COD removal efficiency during the microaeration period increased markedly to 85.37 %, whereas the average sulfate removal efficiency and electron flow ratio decreased to 33.14 % and 7.57 %, respectively. These findings are consistent with those of previous studies demonstrating that microaeration enhances COD removal by suppressing the competition between SRB and substrates (Muller et al., 2022; Ruan et al., 2019). Reactor R2 was operated with microaeration and nitrate addition, whereas reactor R3 was operated with nitrate addition. In reactors R2 and R3, the nitrate concentrations were sequentially increased to 50, 100, and 200 mg NO_3^--N/L on days 1, 76, and 113, respectively, marking the beginning of each operational phase. However, nitrate in R2 and R3 was completely removed throughout the entire experimental period, and no accumulation of nitrite was observed (Fig. S1). As the nitrate concentration increased, the COD removal in both R2 and R3 improved significantly, whereas the sulfate removal and electron flow ratios declined. Under all nitrate concentrations, R2 demonstrated higher COD removal and lower sulfate removal and electron flow ratios than R3. Under typical conditions with nitrate concentrations of 100 and 200 mg NO_3^--N/L, the average COD removal efficiencies of R2 were 95.2 % and 96.9 %, respectively. These values surpassed those of R1 (85.4 %) under microaeration alone between days 136 and 166, and R3 (83.1 %) at 200 mg NO_3^--N/L from days 113–166. Although the COD removal efficiencies of R2 were comparable under both 100 and 200 mg NO_3^--N/L conditions, average sulfate removal was 59.9 % at 100 mg NO_3^--N/L, markedly higher than the 10.8 % observed at 200 mg NO_3^--N/L. The increased COD removal in R2 under the 100 mg NO_3^--N/L condition can be attributed to the synergistic effect of microaeration, which promotes the removal of sulfide from the aqueous phase and its subsequent oxidation, thereby mitigating sulfide inhibition on microbial activity (Nguyen and Khanal, 2018). When the nitrate concentration was increased to 200 mg NO_3^--N/L, sulfate reduction was further inhibited. In R2, the average sulfate removal and electron flow ratio decreased to 10.8 % and 2.3 %, respectively, while R3 exhibited corresponding values of 70.9 % and 17.1 %, respectively. These results indicate that both nitrate addition and microaeration suppressed sulfate reduction to varying extents, with the combined effects of microaeration and nitrate addition in R2 producing a more pronounced inhibitory effect on sulfate reduction. Importantly, nitrate was completely removed in both R2 and R3 across all operational phases with influent NO_3^--N concentrations ranging from 50 to 200 mg/L (Fig. S1). No residual nitrate was detected in the effluent, demonstrating that nitrate supplementation at these levels does not result in nitrate accumulation and is unlikely to negatively impact downstream wastewater treatment processes.

The variations in the effluent sulfide concentration and H_2S produced across the different reactors are shown in Fig. 2(g). In the control reactor R1 (operated without inhibitory conditions during the first 135 days), the sulfide concentration remained stable at 15.3 mg/L, with hydrogen sulfide production averaging 519.1 mg/d. Following 31 days of microaeration operation, these values decreased by 81.88 % and 64.26 %, respectively, reaching 2.8 mg/L and 185.5 mg/d. These results indicate that microaeration conditions effectively suppress sulfide formation, which agrees with the results of previous studies (Nguyen and Khanal, 2018; Zhou et al., 2007). The effluent pH in both R2 and R3 exhibited a progressive increase in response to elevated influent nitrate concentrations. This effect was particularly pronounced at nitrate dosages exceeding 100 mg NO_3^--N/L, with pH values in R2 and R3 consistently surpassing those observed in R1 (Fig. S2). The increased pH was primarily attributable to the alkalinity generation during denitrification processes. In reactor R3, the effluent sulfide concentrations exhibited an increasing trend with increasing nitrate concentration. A higher pH environment facilitated the conversion of H_2S to its less toxic ionic form (HS^-), thereby decreasing the concentration of dissolved H_2S and mitigating its inhibitory impact on microbial communities. Therefore, although the sulfide concentration in R3 reached 29.0 mg/L at a nitrate concentration of 200 mg NO_3^--N/L, the COD removal remained above 80 %. In contrast, reactor R2, which combined microaeration and nitrate addition, demonstrated decreased effluent sulfide concentration as nitrate addition increased. This reduction can be attributed to sulfide oxidation to elemental sulfur or sulfate facilitated by SOB (Nguyen and Khanal, 2018). On day 166, when the R2 and R3 operated stably under nitrate concentration of 200 mg NO_3^--N/L, the effluent sulfide concentration in R2 was measured at as low as 0.3 mg/L. H_2S production decreased in both R2 and R3 as the nitrate concentration increased, likely due to the nitrate-promoted enrichment of SoNRB, which inhibited the sulfate reduction process (Cai et al., 2021; Greene et al., 2003). Under identical nitrate conditions, R2 exhibited significantly lower H_2S production than R3, underscoring the enhanced effectiveness of the combined microaeration and nitrate approach in reducing sulfide levels. Specifically, at a nitrate nitrogen concentration of 200 mg NO_3^--N/L, H_2S production in R2 was 23.1 mg/d, representing a 95.5 % reduction compared to R1 under non-microaerated conditions. This pronounced reduction in R2 was likely associated with a higher abundance of SOB (Nguyen and Khanal, 2018).

Methane production in the different reactors is shown in Fig. 2(h). During the initial 135 days, reactor R1 exhibited stable methane production ranging from 0.63 ± 0.052 to 0.65 ± 0.066 L/d. Following the introduction of microaeration, methane production increased significantly, reaching 1.22 ± 0.055 L/d on day 166, which was 1.88 times higher than that in the non-microaerated period. Methane production in reactors R2 and R3 exceeded that of control group R1 under non-microaerated conditions (day 135) and increased with increasing nitrate concentrations. Reactor R2 exhibited higher methane production compared to Reactor R3 (0.86 ± 0.077 to 1.21 ± 0.066 L/d) at the same nitrate concentration, maintaining methane production values between 1.40 ± 0.088 and 1.57 ± 0.096 L/d. This was 2.15–2.42 times the methane production of R1 under non-microaerated condition (day 135), and 1.15 to 1.29 times that of R1 under microaeration alone. These findings highlight the synergistic effect of microaeration and nitrate addition, which significantly enhanced methane production during the anaerobic treatment of SOW. This improvement can be attributed to the combined conditions effectively reducing sulfide concentrations in the system, thereby alleviating the inhibitory effects of sulfides on microbial growth and metabolism and ultimately improving methane production (Muller et al., 2022; Ruan et al., 2019). Based on a reactor working volume of 4.7 L and a hydraulic retention time of 24 h, the baseline methane production was estimated at 134.04–138.30 L per cubic meter of wastewater. In contrast, microaeration combined with nitrate addition (100–200 mg NO_3^--N/L) increased methane yield to 185.65–195.69 L per cubic meter of wastewater. Using the lower heating value of methane (35.88 MJ/m^3), this improvement corresponds to an additional energy recovery of 6.67–7.02 MJ per cubic meter of wastewater. At a DO concentration of 0.20–0.25 mg/L, the electricity consumption associated with microaeration was approximately 0.4 kWh per cubic meter of

wastewater, corresponding to an energy input of 1.44 MJ per cubic meter of wastewater based on unit conversion. Assuming that nitrate was supplied via nitrate-rich wastewater without additional chemical reagent costs, this preliminary assessment indicates that the energy benefits associated with enhanced methane recovery substantially outweigh the energy inputs required for microaeration, while simultaneously achieving the treatment of nitrate-rich wastewater.

3.2. Sludge characteristics

3.2.1. Particle size distribution

Particle size distributions and morphological observations of the sludge are shown in Fig. 3(a) and Fig. S3. Sludge with a particle size greater than 0.2 mm is typically classified as granular sludge (Liu et al., 2020). After 135 days of operation, the proportion of granular sludge in R1 was only 5.5 %. However, when the operating conditions were changed to microaeration, the proportion increased to 10.2 % within 32 days. This finding suggests that microaerobic conditions promote sludge granulation by reducing sulfide production and creating a more favorable environment for microbial growth and metabolic activity. Previous studies have shown that microaeration enhances intracellular electron transfer and metabolic activity, which leads to increased ATP generation and promotes the secretion of EPS (Wu et al., 2024). These EPS act as biological adhesives that facilitate particle aggregation and granulation (Wu et al., 2024). In reactors R2 and R3, the proportion of granular sludge (particles >0.2 mm) was significantly higher, at 55.6 % and 37.7 %, respectively. The enhanced granulation observed in R2 and R3 can be attributed to the suppression of sulfate reduction, which mitigates the inhibitory effects of sulfides on microbial growth. Reactor R2, which was operated under combined microaerobic conditions and nitrate supplementation, exhibited the highest granulation. This indicated that the synergistic effects of microaeration and nitrate addition effectively accelerated sludge granulation. It is also noteworthy that the hydraulic shear generated by the external aeration recirculation system further enhanced sludge granulation by facilitating particle collisions and increasing aggregate compactness (Chen et al., 2020).

3.2.2. Settling property and VSS/TSS ratio

The variations in SVI_{30} and VSS/TSS ratios are shown in Fig. 3(b). The SVI_{30} values in all three reactors decreased progressively, reaching <45 mL/g by Day 166. This result indicated that granular sludge with excellent settling properties had been developed. Among the reactors, R2 exhibited the lowest SVI_{30}, and this is likely because it contained a higher proportion of granular sludge. The VSS/TSS ratio followed a trend of an initial increase and then a gradual decrease over time. During the early operational stages, heightened microbial activity in the sludge led to the accumulation of organic matter, such as cytoplasmic material and metabolic byproducts, thereby increasing the VSS fraction. However, as the operational time progressed, the accumulation of inorganic solids, including salts, likely contributed to an increase in TSS, resulting in a decline in the VSS/TSS ratio (Chen et al., 2019). By day 166, the VSS/TSS ratio in reactors R1, R2, and R3 had stabilized at approximately 0.7, reflecting high metabolic activity in the sludge (Corsino et al., 2017).

3.2.3. EPS

EPS secreted by microorganisms play a crucial role in sludge granulation and serve as a key indicators of microbial metabolism. The variations in the EPS content across the different reactors are shown in Fig. 3(c). Consistent with a previous study (Liang et al., 2019), the EPS content initially increased and subsequently decreased during granulation. During the first 112 d, reactor R2 exhibited a consistently higher EPS content than R1 and R3, indicating that the combined effects of microaeration and nitrate addition created more favorable conditions for microbial EPS secretion. This enhancement may be attributed to two synergistic mechanisms: (1) microaeration may enhance microbial metabolic activity and energy availability, which could promote EPS secretion (Wu et al., 2024); and (2) nitrate addition facilitated the enrichment of EPS-producing denitrifying bacteria such as *Thauera* and *unclassified_f_Rhodocyclaceae*, as identified in Section 3.3. These microorganisms are known to secrete substantial amounts of EPS to enhance biofilm and granule formation under denitrifying conditions. An elevated EPS content not only increases microbial adhesion and

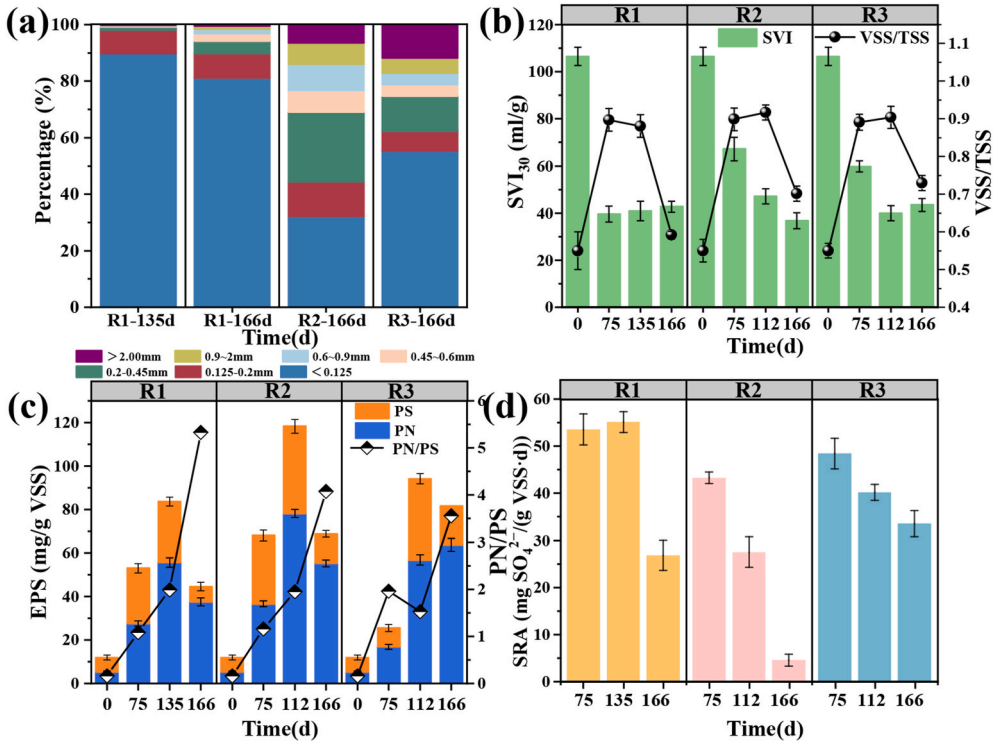


Fig. 3. Sludge characteristics: (a) Particle size distribution; (b) SVI_{30} and the ratio of VSS to TSS; (c) EPS content; (d) Sulfate-reducing activity.

aggregation, promoting the formation and stability of granular sludge (Corsino et al., 2017), but also enhances microbial resistance to sulfide toxicity, thereby improving overall system robustness (Wang et al., 2023b). Throughout the experimental period, the PN/PS ratio remained above 1 and increased, suggesting that PN was the dominant component of sludge EPS. Furthermore, the PN content in R2 consistently surpassed that of R1 and R3. The positively charged amino groups in PN can neutralize the negative charge of sludge, increase sludge hydrophobicity, and reduce the Gibbs free energy, thereby facilitating sludge granulation (Corsino et al., 2017).

3.2.4. Sulfate-reducing activity

The SRA over different time intervals is shown in Fig. 3(d). After a 135-day acclimation period with SOW, the SRA in reactor R1 increased to 53.50 mg SO_4^{2-}/(g VSS·d). The subsequent introduction of microaeration over 31 days led to a significant reduction in SRA, decreasing to 26.78 mg SO_4^{2-}/(g VSS·d). This decline corresponded closely to the sulfate removal trend observed in Fig. 2(b). The reactors R2 and R3 achieved stable operation at nitrate concentrations of 50, 100, and 200 mg NO_3^--N/L on days 75, 112, and 166, respectively. The SRA in both reactors decreased progressively with increasing nitrate concentrations. Reactor R2, which incorporated microaeration, exhibited lower SRA than reactor R3 at equivalent nitrate concentrations. When the nitrate concentration reached 200 mg NO_3^--N/L, the SRA in R2 decreased to 4.48 mg SO_4^{2-}/(g VSS·d) under the synergistic effects of microaeration and nitrate addition. This value represented only 8.4 %, 16.7 %, and 13.4 % of the SRA in R1 under non-microaerated conditions, R1 under microaeration alone, and R3 with nitrate addition, respectively. These findings demonstrate that combining microaeration and nitrate addition effectively suppressed sulfate reduction. This inhibition reduces sulfide toxicity to microorganisms, facilitating improved pollutant conversion and methane production. This explains the enhanced COD removal and methane production observed in R2.

3.3. Microbial community analysis

3.3.1. Microbial richness and diversity

Microbial communities in sludge samples from different reactors were analyzed via high-throughput sequencing, including seeded sludge (S0), reactor R1 without microaeration (R1-135), R1 with microaeration (R1-166), R2 with microaeration and 100 mg NO_3^--N/L nitrate (R2-112), R2 with microaeration and 200 mg NO_3^--N/L nitrate (R2-166), R3 with 100 mg NO_3^--N/L nitrate (R3-112), and R3 with 200 mg NO_3^--N/L nitrate (R3-166). The diversity indices of the samples are represented in Table 1. All samples exhibited coverage exceeding 0.99, indicating that the sequencing results accurately reflected microbial community structures. Higher values of the abundance-based coverage estimator (ACE) and Chao indices indicate increased microbial community richness, while a higher Shannon index reflects greater microbial diversity. The ACE and Chao indices ranked as follows: (R2-112, R2-166) > R1-166 > (R3-112, R3-166) > R1-135. These results indicate that microaeration and nitrate addition enhanced microbial community richness, with the highest richness observed under combined microaeration and nitrate supplementation conditions. A comparison of the Shannon diversity index between R1-135 and R1-166 revealed that microaeration alone slightly reduced the microbial diversity. However, the Shannon indices for reactors R2 and R3 showed an increasing trend with increasing nitrate concentration, with R2 consistently exhibiting higher microbial diversity. This increase in microbial diversity is likely linked to reduced sulfide concentrations, which mitigate their inhibitory effects on microbial proliferation in anaerobic systems.

Table 1
Alpha indices of sludge samples.

Sample	ACE	Chao	Shannon	Coverage
S0	289	289	3.60	1.0000
R1-135	139	139	3.41	1.0000
R1-166	227	227	3.31	1.0000
R2-112	268	266	3.51	0.9998
R2-166	266	264	4.04	0.9999
R3-112	150	150	2.74	1.0000
R3-166	195	193	3.58	0.9998

Non-metric multidimensional scaling analysis (Fig. S4) demonstrated that the S0 sample was clearly separated from the other sludge samples, indicating a substantial shift in microbial community structure after cultivation in SOW. The shortest distance was observed between R1-135 and R1-166, suggesting that microaeration alone exerted a relatively limited effect on the microbial community structure compared to nitrate addition or the combined effects of microaeration and nitrate. The distances between R1-135 and R2-112 or R2-166 were significantly greater, indicating that the interaction between microaeration and nitrate caused pronounced changes in the microbial community structure within the anaerobic reactor. These shifts in the microbial community structure are likely the primary factors underlying the observed differences in organic matter removal efficiency and H_2S production among the reactors.

3.3.2. Microbial community composition

The microbial community composition at the phylum level is shown in Fig. 4(a). The dominant phyla in seed sludge (S0) were Proteobacteria (39.4 %), Desulfobacterota (16.1 %), and Chloroflexi (13.8 %). After 135 d of cultivation with SOW, the dominant phyla in R1 shifted to Proteobacteria (21.2 %), Bacteroidota (23.3 %), Firmicutes (22.7 %), and Desulfobacterota (12.8 %). When R1 was operated under microaeration, the relative abundance of Caldisericota in R1-166 increased to 21.3 %. In contrast, the relative abundances of Proteobacteria and Firmicutes decreased by 15.9 % and 12.8 %, respectively. Sludge samples from R2 (R2-112 and R2-166) and R3 (R3-112 and R3-166) demonstrated higher relative abundances of Chloroflexi, Actinobacteriota, and Hydrogenedentes than those from R1-135. The abundance of these phyla was positively correlated with increasing nitrate concentrations. Previous studies have shown that Chloroflexi and Actinobacteriota contribute to enhanced methane production (de Sousa e Silva et al., 2024). Members of the Chloroflexi family play critical roles in organic matter hydrolysis and nitrogen cycling (Zhang et al., 2021). Moreover, some SOB within Chloroflexi can oxidize sulfur to sulfate (Luo et al., 2020). Actinobacteriota can efficiently remove contaminants from highly concentrated organic wastewater (Zhong et al., 2023), whereas Hydrogenedentes can hydrolyze organic pollutants into acetate (Nobu et al., 2015). The increased abundance of these phyla likely contributed to enhanced organic matter degradation in the nitrate-supplemented reactors. At the same nitrate concentration, reactor R2 exhibited a significantly higher abundance of organic matter-degrading bacteria such as Chloroflexi, Hydrogenedentes, Synergistota, and Armatimonadota, which likely explains the higher COD removal observed in R2. Furthermore, R2-166 showed a greater abundance of Euryarchaeota and Halobacterota than did R1-166, R3-112, and R3-166. These archaeal phyla are known for their role in methanogenesis (Ren et al., 2024). The elevated abundance of methane-producing archaea enhanced the methane production in R2 under the synergistic effects of nitrate addition and microaeration.

The microbial community composition at the genus level is shown in Fig. 4(b). In R1-135, the dominant genera were *Desulfovibrio* (12.8 %), *unclassified_f_Aeromonadaceae* (12.2 %), and *Clostridium_sensu_stricto_6* (9.3 %). Following microaeration, the relative abundances of these genera decreased by 2.5 %, 11.7 %, and 8.3 %, respectively. Conversely, the relative abundance of *Caldisericum*, known for degrading organic matter into acetate and hydrogen to promote methane production (Wang et al., 2023a), significantly increased from 7.14 % to 28.47 % in R1-166. *Desulfovibrio*, a dissimilatory SRB capable of reducing sulfate to sulfide (Fadhlaoui et al., 2018), exhibited reduced relative abundances in R2-112, R2-166, R3-112, and R3-166 by 12.08 %, 12.76 %, 8.88 %,

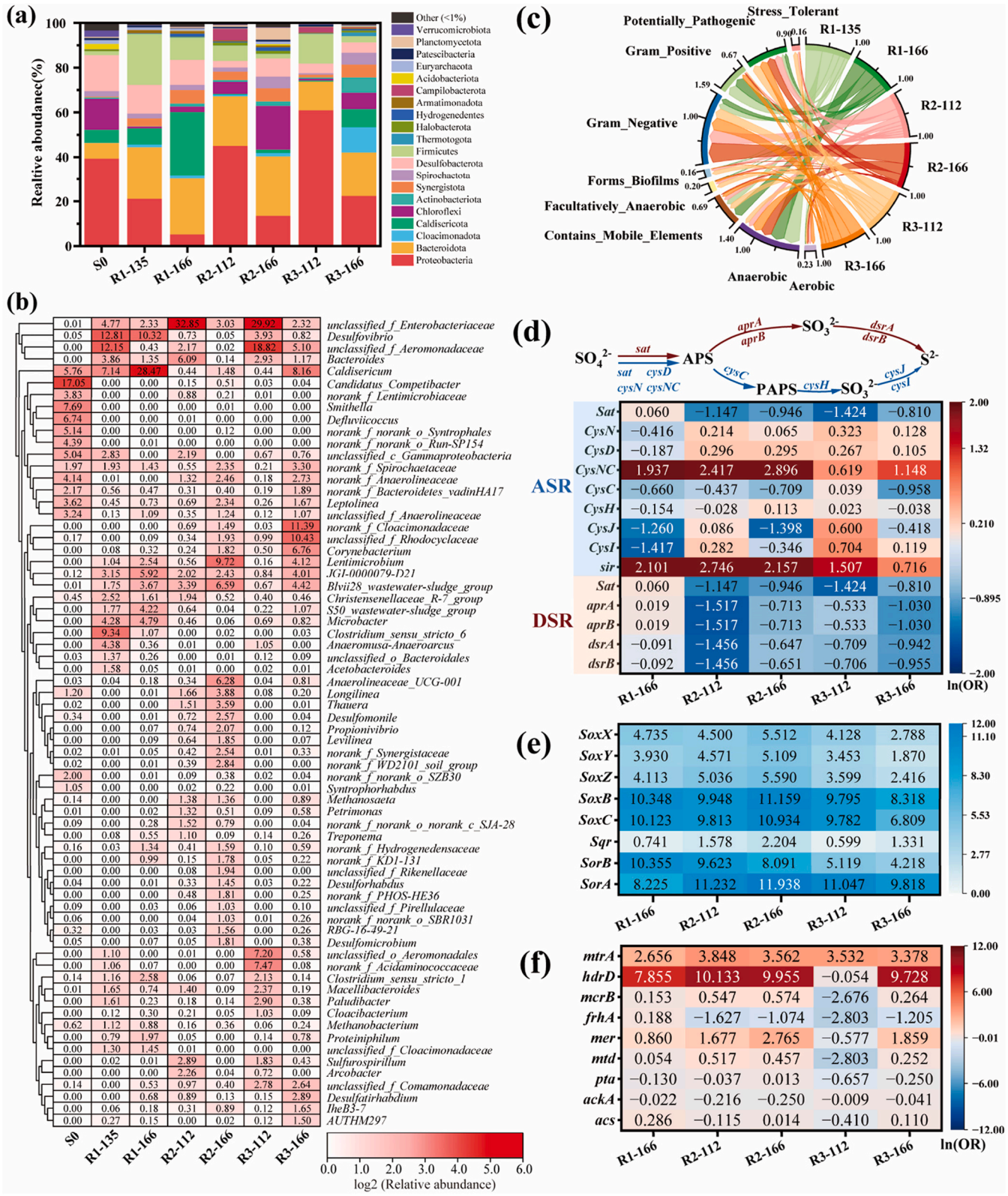


Fig. 4. Microbial community structure and function analysis. (a) Microbial community composition at phylum level; (b) Microbial community composition at genus level; (c) Microbial phenotypes distribution; (d) The odd ratio of sulfate reduction genes relative to the control; (e) The odd ratio of sulfur oxidation genes relative to the control; (f) The odd ratio of methanogenesis-related genes relative to the control; (The phylum/genera with relative abundance less than 1 % were categorized as "other").

and 11.99 %, respectively, compared to R1-135. These reductions were more pronounced than those observed in R1-166, indicating that nitrate addition, particularly when combined with microaeration, significantly inhibited SRB growth. This phenomenon explains the lower H_2S production observed in R2. Compared to R1-135 and R1-166, sludge samples from R2 (R2-112 and R2-166) and R3 (R3-112 and R3-166) exhibited higher relative abundances of *norank_f_Cloacimonadaceae*, *unclassified_f_Rhodocyclaceae*, and *Corynebacterium*, and their abundances increased with nitrate concentration. *norank_f_Cloacimonadaceae* is an acetic acid-producing bacterium that converts propionic acid to acetate (Hu et al., 2022a). Both *unclassified_f_Rhodocyclaceae* and *Corynebacterium* are denitrifying bacteria (Negi et al., 2016; Si et al., 2023), which can inhibit SRB growth through substrate competition and nitrite formation during denitrification (Xu et al., 2017). Additionally, the sludge samples from R2 and R3 showed higher abundances of acetotrophic methanogens (*Methanosaeta*) and hydrogenotrophic methanogens (*Methanobacterium*) than those from the control group R1-135. Enrichment of methanogens enhanced methane production in R2 and R3. R2-166 exhibited a higher abundance of *Blvii28_wastewater-sludge_group*, *Lentimicrobium*, *Longilinea*, *Thauera*, and *Desulfomicrobium* than the other samples. *Blvii28_wastewater-sludge_group* is widespread in wastewater treatment systems and plays an important role in organic matter degradation and denitrification (Li et al., 2023). A previous study has shown that anaerobic digestion reactors with a higher abundance of *Blvii28_wastewater-sludge_group* typically have higher COD removal and methane production (Chen et al., 2023b). *Lentimicrobium* and *Longilinea* are fermentative bacteria that convert organic matter to volatile fatty acids (Huang et al., 2022; Zhang et al., 2017). *Desulfomicrobium* degrades various organic pollutants (Yang et al., 2024). *Thauera*, a known SoNRB, can oxidize S^{2-}, HS^- and H_2S to elemental sulfur or SO_4^{2-} at rates 20 to 30 times faster than the reduction of SO_4^{2-} or elemental sulfur (Jiang et al., 2009). Rapid sulfur oxidation reduces sulfide accumulation and facilitates the conversion of pollutants to methane, thereby enhancing both COD removal and methane production in anaerobic systems. Enriching these genera in R2-166 underscores their superior performance in pollutant degradation and methane production.

3.3.3. Function annotation

The distribution of microbial phenotypes is shown in Fig. 4(c). In R1-135, the relative abundance of anaerobic microbes was 37.18 %. The microaerobic environment facilitated the proliferation of facultatively anaerobic bacteria, as reflected by a 4.86 % reduction in anaerobic microbes and a slight increase in facultatively anaerobic microbes of 0.05 %. Facultative fermenters, with their rapid oxygen consumption capacity, are likely to deplete dissolved oxygen, thereby creating protective niches for anaerobic organisms (Kato et al., 1993; Zhu et al., 2009). Bacteria associated with biofilm formation and stress tolerance were more abundant in R2 and R3 at a nitrate concentration of 100 mg NO_3^--N/L. These findings suggest that nitrate and synergistic microaerobic conditions promote microbial aggregation and enhance resilience to environmental stress. However, when the nitrate concentration increased to 200 mg NO_3^--N/L, the relative abundance of these bacteria decreased. The dynamics of biofilm-forming microbes correspond to changes in the EPS content during sludge granulation. Moreover, the reduced abundance of stress-tolerant microbes at 200 mg NO_3^--N/L nitrate may be attributed to suppressed sulfide production under these conditions.

PICRUSt2 was employed to predict functional genes potentially involved in sulfate reduction, sulfur oxidation, and methanogenesis. The results are shown in Fig. 4(d–f). The sulfate-reducing metabolism involves both assimilatory sulfate reduction (ASR) and dissimilatory sulfate reduction (DSR). The first step in both pathways is the conversion of sulfate to adenosine-5′-phosphosulfate (APS), mediated by the key gene *sat*. Compared to R1-135, *sat* was downregulated in both R2 and R3, indicating that nitrate addition, either alone or in combination with microaeration, significantly inhibited the conversion of sulfate to APS (Cai et al., 2021). This trend corresponds to a marked reduction in the SRA under these conditions. Genes involved in the ASR, including *cysC*, *cysJ*, and *cysI*, were downregulated in sludge samples from reactors R2 and R3, with the most pronounced decrease observed in R2-166. In the DSR metabolic pathway, *dsrA* and *dsrB* showed a decline in R1-166, whereas *sat*, *aprA*, and *aprB* were upregulated. However, the DSR-related genes (*sat*, *aprA*, *aprB*, *dsrA*, and *dsrB*) were significantly down-regulated in R2 (R2-135 and R2-166) and R3 (R2-135 and R2-166). These findings suggest that microaeration and nitrate addition inhibit sulfate reduction through different mechanisms. Specifically, in R3, the abundance of DSR-related genes decreased with increasing nitrate concentration, likely because of the competitive advantage of NRB over SRB for substrates (Xu et al., 2017). In contrast, under the combined nitrate and microaeration conditions, DSR-related genes in R2 showed a slight increase as nitrate concentrations increased. This could be attributed to the enhanced pollutant conversion, generating more organic acids, which may mitigate substrate competition between NRB and SRB, thereby slightly promoting the SRB profile. This phenomenon can be attributed to the enhanced conversion of pollutants, generating more organic acids. These organic acids may alleviate substrate competition between NRB and SRB, thereby modestly favoring SRB activity. Compared with the control (R1-135), genes associated with sulfur oxidation were upregulated under microaeration, nitrate addition, and their combination (Fig. 4(e)), suggesting an enrichment of SOB in the anaerobic digestion system. Furthermore, the sulfur oxidation genes *SoxX*, *SoxY*, *SoxZ*, *SoxB*, *SoxC*, and *SorA* exhibited higher abundance under the synergistic effects of nitrate and microaeration. This may account for the lower sulfide concentrations and reduced hydrogen sulfide production observed in R2. Key genes involved in methanogenesis, such as *mtrA*, *hdrD,* and *mer*, which mediate acetate fermentation and CO_2 reduction to methane (Liu et al., 2019), were significantly upregulated in R2 (Fig. 4(f)). This indicates that the combination of microaeration and nitrate addition effectively enhanced methane metabolic activity, consistent with the elevated methane production observed in reactor R2. It should be emphasized that the PICRUSt2 analysis provides predicted functional potentials rather than direct measurements of gene expression, representing possible rather than actual metabolic activities of microbial communities. Therefore, these predictive results should be interpreted with caution and regarded as preliminary indications that require further validation through metagenomic or metatranscriptomic sequencing.

3.4. Mechanisms of microaeration and nitrate synergy in enhancing SOW treatment

The synergistic mechanisms through which nitrate and microaerobic conditions enhance the anaerobic digestion of SOW were elucidated based on the sludge characteristics and microbial community composition (Fig. 5). External circulation aeration facilitated the volatilization of a small portion of H_2S. Based on a static gas-liquid equilibrium assumption, the concentrations of dissolved H_2S in reactors R1, R2, and R3 on day 166 were calculated to be 0.24, 0.01, and 3.98 mg/L, respectively, by accounting for total sulfide concentration, system pH, and the dissociation equilibrium of H_2S. Accordingly, H_2S volatilization associated with aeration was estimated to account for only approximately 0.19 %, 0.0077 %, and 3.08 % of the total influent sulfur per day in R1, R2, and R3, respectively. These findings suggest that H_2S volatilization was minimal and can be considered negligible in the overall sulfur balance. Conversely, the introduction of microaeration-enriched fermentative bacteria, such as *Lentimicrobium*, *Longilinea*, *norank_f_Cloacimonadaceae*, and *Blvii28_wastewater-sludge_group*, enhanced the hydrolysis and fermentation of organic matter into VFAs (Hu et al., 2022a; Huang et al., 2022; Li et al., 2023; Zhang et al., 2017). Moreover, the combined effects of nitrate and microaeration promoted the proliferation of SOB, including *Sulfurospirillum* and *Thauera*, leading to significant upregulation of sulfur oxidation-related genes such as *SoxB*,

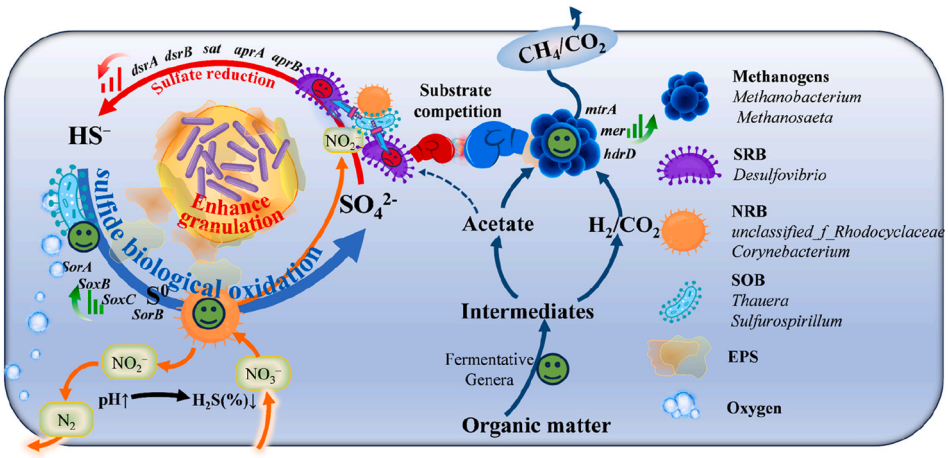


Fig. 5. The proposed mechanism of nitrate and microaeration synergy in enhancing anaerobic digestion of sulfate-containing organic wastewater.

SoxC, *SorA*. SOB oxidize sulfides into elemental sulfur or sulfate, alleviating the inhibitory effects of sulfides on microbial activity and particularly benefiting methanogens (Jiang et al., 2009). Additionally, nitrate supports the growth of heterotrophic NRB, such as *unclassified_f_Rhodocyclaceae* and *Corynebacterium*, which can inhibit SRB growth through substrate competition (Negi et al., 2016; Si et al., 2023; Xu et al., 2017). During nitrate reduction, heterotrophic NRB produced nitrite and alkalinity, resulting in an increase in the pH of the wastewater to 7.9–8.4. This pH elevation shifted the equilibrium of H_2S toward less toxic forms such as HS^- and S^{2-}. Nitrite concentrations in the effluent of reactors R2 and R3 were consistently below 0.34 mg/L (as shown in Fig. S1 of the Supplementary Information), indicating no significant accumulation of nitrite. Despite this, nitrite is a well-known transient intermediate in the denitrification process. Previous studies have demonstrated that nitrite can inhibit SRB growth by disrupting cell membranes and affecting periplasmic and cytochrome systems (Greene et al., 2003; Xu et al., 2017). The presence of nitrate also suppresses SRB through competition for electron donors and carbon sources, as evidenced by the downregulation of genes associated with dissimilatory sulfate reduction (*sat*, *aprA*, *aprB*, *dsrA*, and *dsrB*) (Lai et al., 2020). Furthermore, nitrate-microaeration conditions increased the abundance of EPS-secreting bacteria such as *Thauera* and *unclassified_f_Rhodocyclaceae*, which facilitated sludge granulation (Li et al., 2024b). The formation of granular sludge improves the metabolic transfer of intermediates between fermentative bacteria and methanogens while enhancing the resilience of the microbial community to sulfides (Xue et al., 2017). Consequently, reactors operating under nitrate-microaerobic conditions exhibited higher COD removal efficiencies and methane production, with the combined strategy improving COD removal efficiency and methane production by 48.2–49.8 % and 138.5–141.5 %, respectively.

3.5. Proposal of a practical application strategy

This study demonstrated that, compared to the control reactor, which was operated without microaeration or nitrate addition, maintaining a DO concentration of 0.20–0.25 mg/L and adding 100–200 mg NO_3^--N/L of nitrate led to a reduction in H_2S production by 79.2–95.5 % while improving COD removal efficiency by 49.1–50.8 % and methane production by 138.5–141.5 %. These observations were derived from laboratory-scale experiments using glucose as the primary carbon source, which was deliberately selected to facilitate mechanistic investigation under well-controlled conditions. Based on these findings, a potential strategy is proposed for the treatment of LCDA wastewater from petrochemical enterprises (Fig. 6). In this conceptual process, maintaining a DO concentration of 0.20–0.25 mg/L, together with the addition of 100–200 NO_3^--N mg/L nitrate, is expected to significantly reduce H_2S production, enhance COD removal efficiency, and increase methane production, as demonstrated in our laboratory-scale experiments. The nitrate nitrogen required for this process could be sourced from refinery flue gas denitrification wastewater.

LCDA wastewater, generated from the microbial fermentation of n-alkanes (C10–C18), is mainly composed of sulfate, dicarboxylates, and fermentation medium components, among which glucose constitutes the dominant biodegradable carbon substrate. It is characterized by a low pH (3.0–4.5) and a high concentration of COD (>6000 mg/L), sulfate

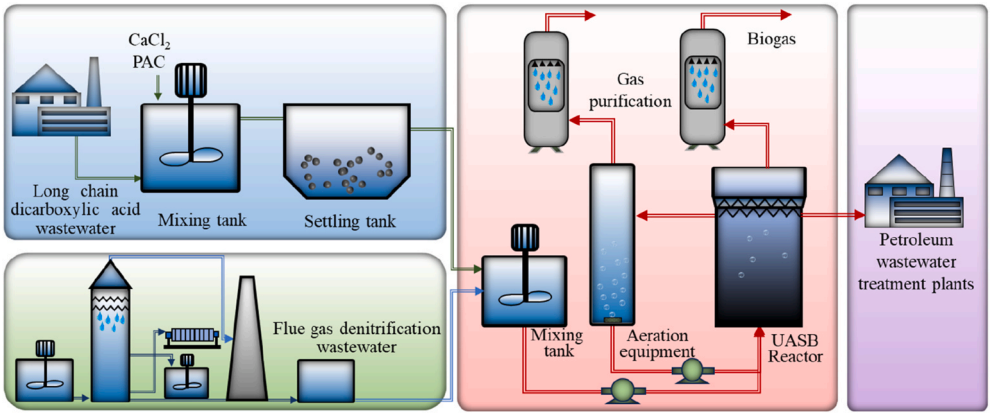


Fig. 6. Combined treatment process for long-chain dicarboxylic acid (LCDA) wastewater and flue gas denitrification wastewater.

(>10000 mg/L), total phosphorus (>272 mg/L), and ammonium nitrogen (NH_4^+-N: 17.91–89.34 mg/L), classifying it as typical SOW. Given that glucose-based substrates dominate the organic fraction of LCDA wastewater, the synthetic wastewater employed in this study provides a representative basis for investigating the key microbial interactions and sulfur transformation processes relevant to LCDA treatment, although the practical wastewater exhibits higher compositional complexity. Drawing on previous studies (Chen et al., 2023a), we suggest that pretreatment with 15 g/L $CaCl_2$ and 20 g/L polyaluminum chloride (PAC) could effectively remove total phosphorus and sulfate through precipitation and coagulation processes. Specifically, calcium ions react with phosphate ions to form $Ca_3(PO_4)_2$ and $Ca_{10}(PO_4)_6(OH)_2$, while PAC hydrolysis products (Ala) and sulfate ions co-precipitate as $Ca_4Al_6O_{12}SO_4$. The resulting solid precipitates could be repurposed as raw materials for calcium sulfoaluminate cement production, supporting resource recovery. In addition, small organic molecules such as organic acids, alcohols, and amines can be removed through adsorption, coagulation, and co-precipitation, significantly reducing COD and other pollutants. Following pretreatment, the LCDA wastewater would be mixed with nitrate-rich refinery flue gas denitrification wastewater at a 1:1 volumetric ratio, and the mixture would be introduced into a UASB reactor equipped with an external aeration device. Maintaining a recirculation ratio of 200 % and a DO level of 0.20–0.25 mg/L is anticipated to optimize the biological processes, based on the operational conditions that yielded stable performance and synergistic sulfur control in the present study. The gases released during aeration could be scrubbed with NaOH prior to discharge, and the methane generated could be purified and utilized for energy recovery within the refinery. The final effluent is expected to meet the influent quality standards of petroleum wastewater treatment plants. This integrated treatment strategy is designed to enable the co-management of LCDA and nitrate-rich refinery wastewater, supporting circular resource utilization by reusing waste-derived nitrate and advancing sustainable wastewater management through "waste-to-treat-waste" approach. It should be emphasized that this strategy represents a proposed conceptual design, intended to illustrate the potential application of our research findings. To date, no pilot- or full-scale experiments have been conducted to validate this process. Further research using real LCDA wastewater is therefore required to assess the long-term stability, robustness, and engineering feasibility of the proposed strategy under practical operating conditions.

4. Conclusions

In this study, we systematically investigated the influence of different nitrate concentrations under microaerobic conditions on enhancing anaerobic digestion in SOW. The combined application of nitrate (100–200 mg NO_3^--N/L) and microaeration (DO: 0.20–0.25 mg/L) synergistically promoted EPS secretion and granular sludge formation. This operational strategy effectively inhibited the proliferation of SRB (*Desulfovibrio*) while selectively enriching SOB (e.g., *Sulfurospirillum* and *Thauera*) and methanogens (e.g., *Methanosaeta* and *Methanobacterium*). Sulfate reduction genes (*aprA*, *aprB*, *dsrA*, and *dsrB*) were downregulated, whereas sulfur oxidation genes (*SoxX*, *SoxY*, *SoxZ*, *SoxB*, *SoxC*, and *SorA*) and methanogenesis-related genes (*mtrA*, *hdrD*, and *mer*) were upregulated. The suppression of sulfide production alleviated its inhibitory effects on anaerobic digestion, significantly improving COD removal efficiency (48.2–49.8 %) and methane production (138.5–141.5 %). These findings provide mechanistic insights into the interactions between microaeration and nitrate addition, offering a promising strategy for optimizing the anaerobic treatment of SOW and controlling sulfide production. However, future work should aim to validate these results using multi-omics and real-world applications to further refine the practical viability of this approach.

CRediT authorship contribution statement

Jiazheng Ding: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis. **Jiahao Liang:** Writing – original draft, Visualization, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Changgang Li:** Writing – review & editing, Visualization, Project administration, Methodology, Investigation, Formal analysis. **Qinghong Wang:** Writing – review & editing, Funding acquisition. **Wenyu Xie:** Writing – review & editing, Project administration, Funding acquisition. **Xiang Tan:** Visualization, Investigation. **Chunmao Chen:** Writing – review & editing. **Dehao Li:** Writing – review & editing, Project administration, Funding acquisition. **Yali Zhan:** Writing – review & editing. **Xiaoqing Chen:** Writing – review & editing, Funding acquisition. **Wen Fu:** Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jclepro.2026.147650.

Data availability

Data will be made available on request.

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